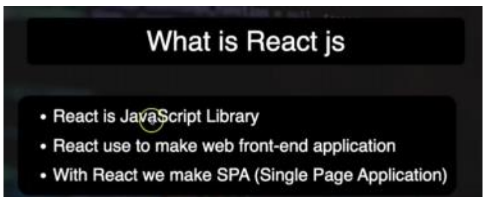
React Js

Wednesday, December 3, 2025 10:46 PM

Install node js
Visual studio
Npm create vite
Npm install
Npm run dev

[React JS 19 Full Course in Hindi | Learn React from Scratch (2025) in One Video](#)

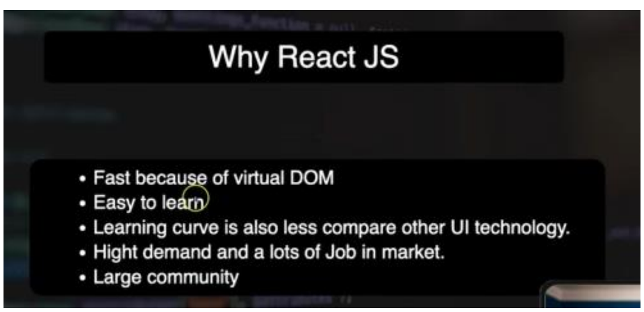
https://vite.dev/guide/
https://react.dev/learn



Section 1:- Introduction to React JS

React js is not framework while angular is framework

Single page application : when we go from one page to another application will not refresh



Section 2:- react js setup with vite tool

Note :
Once we install node js , npm also will get installed
Using npm we can run vite command
Vite will help to create react js project

Vite is a **next-generation build tool** that is often used when creating React applications. It is designed to be **faster**, **lighter**, and **more efficient** than older tools like **Create React App (CRA)** or Webpack.

🚀 **What Vite Does in React**
When you create a React app using Vite, it provides:
1. Super-fast development server
- Vite uses **native ES modules (ESM)** in the browser.
- It starts the dev server almost **instantly**, even for large projects.

2. Lightning-fast Hot Module Replacement (HMR)
- When you edit files, Vite updates the browser **immediately** without refreshing the page.

3. Modern build process
- Uses **Rollup** under the hood for optimized production builds.
- Produces smaller, faster, and more efficient bundles.

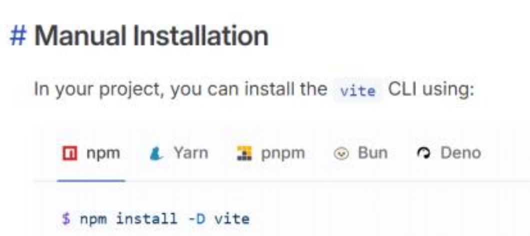
LTS : long term support

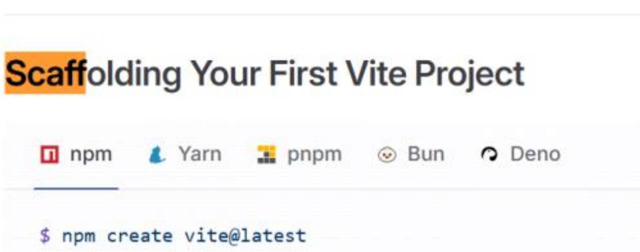
```
C:\Users\Admin>node -v
v24.11.1

C:\Users\Admin>npm -v
11.6.2
```

Vite is use to set up react js (use vite or parcel)

Frontend development tool
https://vite.dev/guide/





SWC: speedy web compiler

Command : to create project

Npm create vite

```
  Shortcuts
  press r + enter to restart the server
  press u + enter to show server url
  press o + enter to open in browser
  press c + enter to clear console
  press q + enter to quit
uuu
u

  ➜  Local:   http://localhost:5173/
  ➜  Network: use --host to expose
```

☑ **What npm install Does**

When you run:

npm install
it:
1. Installs all dependencies listed in package.json
Example fields:

```
"dependencies": {
  "react": "^18.2.0",
  "react-dom": "^18.2.0"
},
"devDependencies": {
  "vite": "^5.0.0"
}
```
npm install downloads all these packages.

2. Creates the node_modules folder
This folder contains all installed packages + their internal dependencies (thousands of small files).

3. Creates or updates package-lock.json
This file ensures all systems get the exact same versions of packages.

4. Rebuilds dependencies if needed
Sometimes npm needs to compile native modules.

🎯 **In simple words:**
npm install downloads all the libraries your project needs so your app can run.

Step to create project

Npm create vite
Npm install (will install all dependencies and install node modules)
Npm run dev

This is latest ----------------imp
npm create vite@latest my-react-app -- --template react

Go inside my-react-app :- npm install
Npm run dev
http://localhost:5173/

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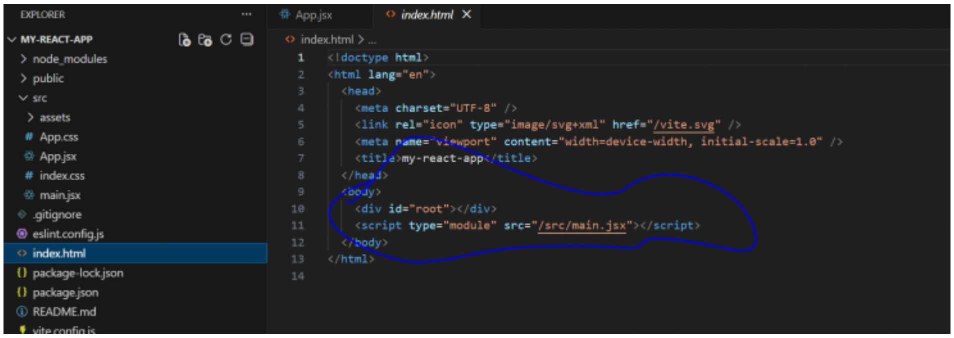
Section 3:- first code or hello world in react js

Ctrl+shift+d
It will ask to create launch.json for debug

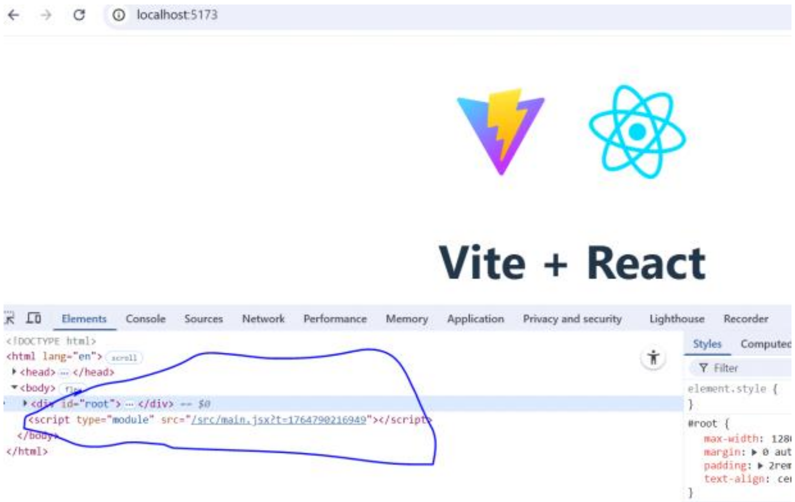
```
{
  "version": "0.2.0",
  "configurations": [
    {
      "name": "Debug React + Vite",
      "type": "pwa-chrome",
      "request": "launch",
      "url": "http://localhost:5173",
      "webRoot": "${workspaceFolder}\\src",
      "breakOnLoad": true
    }
  ]
}
```

Imp :
Step 1 :-Index.html execute or load first in project

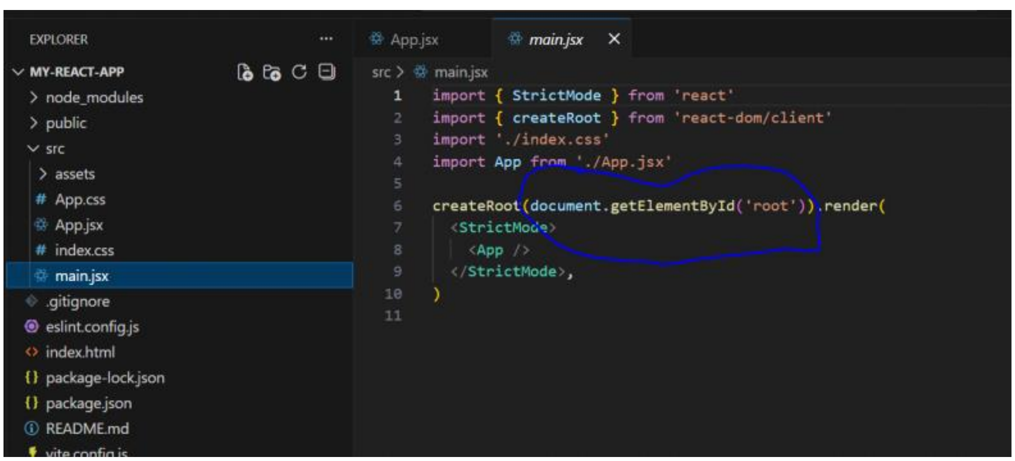
All the code will inside div root



all code will be under div root



Step 2:- Script tag will load main.jsx file



Using document.getElementById('root')------------ it will search root in index.html whatever **App.jsx code** is there
That will inculde in **root div and display**

Main.jsx :- include app.jsx file

Note :
app.jsx will called as **component**

In main.jsx :--./index.css which represent css is applied

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For understanding

1. What is JSX?
JSX stands for **JavaScript XML**. It’s a syntax extension for JavaScript that lets you **write HTML-like code inside JavaScript**.
Example:

```
function App() {
  return (
```

```
    <div>
      <h1>Hello, React!</h1>
    </div>
  );
}
```
Here:
- <div> and <h1> look like HTML.
- But this code is actually JavaScript under the hood.
- React transforms JSX into React.createElement() calls when building.

2. Why .jsx instead of .js?
- **Clarity:** .jsx tells tools and developers that the file contains JSX syntax.
- **Tooling:** Some editors, linters, and compilers treat .jsx files differently, enabling syntax highlighting, autocomplete, and error chec king for JSX.
- **Optional:** Technically, React can handle JSX inside .js files too, but using .jsx is considered a best practice.

3. How it works with Vite or React
- Vite (or Babel, CRA, Next.js, etc.) **automatically compiles JSX** into regular JavaScript that browsers understand.
- Example compiled code from JSX:

```
// JSX
const element = <h1>Hello</h1>;
// Compiled JS
const element = React.createElement('h1', null, 'Hello');
```

✅ **In short:**
- .jsx = JavaScript + HTML-like JSX code
- Helps **React developers** write components in a clean, readable way
- Compiled into standard JavaScript by build tools like **Vite**

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5. Copilot Shortcuts

Action	Shortcut (Windows)
Trigger suggestions	Ctrl+Enter
Accept suggestion	Tab
Show alternative	Alt+[/ Alt+]
Open Copilot panel	Ctrl+Shift+P → "Copilot: Open Panel"

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1. What export does
- export is used to make a variable, function, or class available to **other modules/files**.
- Without export, the code is private to that file.

2. What default means
- default means this is the **main thing exported from this file**.
- When another file imports this module, it can import it **without using curly braces**.

3. Example in React
Suppose you have a React component App.js:

```
function App() {
   return (
      <div>
         <h1>Hello, World!</h1>
      </div>
   );
}
// Export the component as the default export
export default App;
```
Now, in another file, you can import it like this:

```
import App from './App';  // No curly braces needed
function Main() {
   return <App />;
}
```
- If it were a **named export** (export { App };), you would need curly braces:

```
import { App } from './App';
```

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Option A: Temporarily Disable

- Press **Ctrl+Shift+P** (Windows) to open the Command Palette.
- Type **“Copilot: Disable”**.
- Select:
  - **“GitHub Copilot: Disable for this workspace”** → stops suggestions only in the current project.
  - **“GitHub Copilot: Disable globally”** → stops suggestions in all projects.

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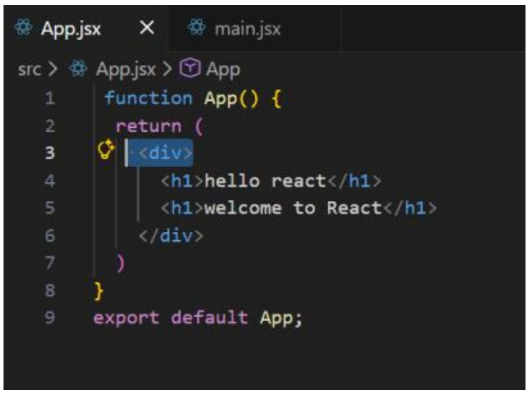
Note :
Html tag start with small element and react component should be start with cap then only
It understand this is react tag

Always create component with capital letter (function)

If we are using multiple h1 we need to wrap it in div tag



JSX : javascript xml

Library flow we can decide and framework is having own flow

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Section 4:- Use and Learn React without installation